

Remark: The following questions don't appear numbered in the scaling book. I will number them in order of appearance as 0.X where $X \in \mathbb{N}$ and $X \geq 1$.

Question 0.1

Per token KV cache size is:

$$\frac{2 \cdot S \cdot L \cdot K \cdot H}{S} = 2 \cdot L \cdot K \cdot H$$

where S is sequence length. Hence in int8:

$$2 \cdot 80 \cdot 8 \cdot 128 = 163840 \text{ Bytes} \simeq 164 \text{ KB}$$

Question 0.2

let M be the number of parameters of the model, let $B = 32$ and $S = 8192$. Assume int8:

$$\begin{aligned} \text{Total Memory} &= M + B \cdot S \cdot 2 \cdot L \cdot K \cdot H = 70e9 + B \cdot S \cdot 164e3 = \\ &\simeq 113 \text{ GB.} \end{aligned}$$

Assuming we serve this model in TPU v5e, the smallest possible slice is a 4x2 since it gives us 128 GB of aggregated HBM.

Question 0.3

At $B = 32$ the MLP block is not compute bound. Therefore using the general equation from chapter 6:

$$\text{Step Time} = \frac{B \times \text{KV cache size}}{BW \cdot N} + \frac{M}{BW \cdot N}$$

where N is the total number of devices in the slice, hence:

$$\text{Step Time} = \frac{32 \cdot 8142 \cdot 164e3}{8.2e11 \cdot 8} + \frac{70e9}{8.2e11 \cdot 8}$$

$$\simeq \frac{113e9}{8.2e11 \cdot 8} \simeq 0.017 \text{ s} = 17 \text{ ms}$$

$$\begin{aligned} \text{Tokens/s/chip} &= B \cdot \frac{1}{0.017} \cdot \frac{1}{N} = \frac{B}{0.017 \cdot N} \\ &= \frac{32}{0.017 \cdot 8} = 235 \text{ Tokens/s/chip.} \end{aligned}$$

If instead we use a 4×4 slice we get:

$$\text{Step Time} = \frac{113 \text{ e } 9}{8.2 \text{ e } 11 \cdot 16} \approx 0.008 \text{ s} = 8 \text{ ms}$$

$$\text{Tokens/s/chip} = \frac{32}{0.008 \cdot 16} = 250 \text{ Tokens/s/chip.}$$

Question 0.4

For bf16 weights and activations:

$$\text{Intensity (Matmul)} = B \geq \frac{1.97 \text{ e } 14}{8.2 \text{ e } 11} \approx 240$$

If int8 weights but FLOPS in bf16, note that

$$\text{Intensity (Matmul)} = \frac{2 \text{ BDF}}{\text{BD} + \text{DF} + \text{BF}} = 2 \cdot \frac{\text{BDF}}{\text{BD} + \text{DF} + \text{BF}}$$

$$\approx 2 \cdot B \geq 240 \Leftrightarrow B \geq 120$$

If int8 weights and int8 flops:

$$\text{Intensity (Matmul)} = 2B \geq \frac{2 \cdot 1.47e14}{8.2e11}$$

$$\Rightarrow B \geq 240$$

Question 0.5

Assume $B = 1$.

For bfloat16:

$$\begin{aligned} \text{Total Memory} &= 2 \cdot M + 2 \cdot B \cdot S \cdot 2 \cdot L \cdot K \cdot H \\ &= 2 \cdot 70e9 + 2 \cdot 8192 \cdot 164e3 \simeq 143 \text{ GB} \end{aligned}$$

so we need at least a 4x4 TPU v5e slice.

For int8:

$$\begin{aligned} \text{Total Memory} &= M + B \cdot S \cdot 2 \cdot L \cdot K \cdot H = 70e9 + 8192 \cdot 164e3 = \\ &\simeq 71 \text{ GB} \end{aligned}$$

so we need at least a 4x2 TPU v5e slice.

For int 4:

$$\text{Total Memory} = \frac{M}{2} + \frac{B \cdot S \cdot 2 \cdot L \cdot K \cdot H}{2} \approx 35.5 \text{ GB}$$

so we need at least a 2×2 TPV use slice.

Question 0.6

For bfloat 16 in a 4×4 slice:

$$\frac{2 \cdot M + 2 \cdot B \cdot S \cdot 2 \cdot L \cdot K \cdot H}{4 \times 4} \leq 16 \text{ GB}$$

note that B cannot be $B7/240$ for otherwise our KV cache size will not fit in this topology. Hence we are memory bound, and choosing the max B that satisfies the conditions just makes us closer to our upper bound, therefore:

$$\text{bfloat 16 Step Time} = \frac{16e9}{8.2e11} \approx 0.019s$$

$$\approx 19 \text{ ms.}$$

For int 8, B cannot be $B \geq 240$, hence as in the bfloat16 case:

int 8 step time with max B = 19 ms.

For int 4, the same argument as above holds, therefore:

int 4 step time with max B = 19 ms.

Question 0.7.

For each case we can use the condition we derived in Exercise 2 of chapter 7 to obtain the max B that fits in each topology.

For int 8 and a 2x4 slice:

$$B \leq \frac{|X| \cdot HAM - M}{2 \cdot S \cdot L \cdot K \cdot H} = \frac{8 \cdot 16 - 17024}{8192 \cdot 16423} \approx 42$$

For bfloat16 and a 4x4 slice:

$$B \leq \frac{1 \times 1 \cdot \text{HBM} - 2 \cdot M}{2 \cdot S \cdot 2 \cdot L \cdot K \cdot H} \simeq 43$$

For int4 and a 2x2 slice:

$$B \leq \frac{1 \times 1 \cdot \text{HBM} - \frac{M}{2}}{\frac{S \cdot 2 \cdot L \cdot K \cdot H}{2}} \simeq 43$$

Therefore:

$$\text{Queries/chip bfloat16 } 4 \times 4 \text{ slice} = \frac{1}{0.014 \cdot 512} \cdot 42 \cdot \frac{1}{16} \\ \simeq 0.26$$

$$\text{Queries/chip int8 } 4 \times 2 \text{ slice} = \frac{1}{0.014 \cdot 512} \cdot 42 \cdot \frac{1}{8} \\ \simeq 0.53$$

$$\text{Queries/chip int4 } 2 \times 2 \text{ slice} = \frac{1}{0.014 \cdot 512} \cdot 42 \cdot \frac{1}{4} \simeq 1.07$$

Question 0.8

If we double our topology, we can see that

$$B \leq \frac{2 \cdot |X| \cdot \text{HBM} - M}{S \cdot 2 \cdot L \cdot K \cdot H} \approx 138$$

for all cases. Therefore since $\frac{138}{42} \approx 3.3$, we 1.65x our peak throughput in each case.

Question 0.9

The only sharding option we have is model parallelism. From chapter 7 we know that we will not be communication bound if:

$$\frac{2 \cdot D \cdot F}{N \cdot W_{\text{HBM}}} > \frac{2 \cdot B \cdot D}{2 \cdot W_{\text{ICI}}} \Leftrightarrow \frac{2F}{(B \cdot \beta)} > N$$

where $\beta = \frac{W_{\text{HBM}}}{W_{\text{ICI}}}$.

Therefore four setting:

$$N \geq \frac{2 \cdot F}{138 \cdot 8} \approx 52$$

so we can avoid being communication bound since $N=32$ in a 4×8 TPU v5e slice.

The KV cache will be sharded along the head dimension and among the batch dimension as well.

$$KV[B_x, L, S, L, K_y, H]$$

where Mesh = $\{x: 4, y: 8\}$.

Question 0.10

We can use the formula:

$$\text{Total FLOPs} = 2 \cdot M \cdot B$$

hence:

$$t = \frac{2 \cdot M \cdot B}{16} \cdot \frac{1}{C \cdot 0.4} \approx 0.9$$

where we assumed bfloat16 FLOPs.

Question 0.11

At batch size 32 we are memory bound during decoding, therefore :

$$\begin{aligned}\text{Step Time} &= \frac{32 \cdot 8192 \cdot 164e3 + 170e4}{N \cdot \text{WMM}} \\ &\simeq 0.008 \text{ s} = 8 \text{ ms.}\end{aligned}$$

$$\text{Sequence Time} = \text{Step Time} * 4096 \simeq 35 \text{ s}$$

$$\text{Decodings Per Second} = \frac{1}{\text{Step Time} \cdot 4096} \cdot 32 \simeq 0.9$$

$$\text{Prefills Per Second} = \frac{1}{0.9} \simeq 1.1$$

$$\text{Per prefill step I do } \frac{0.9 \cdot 32}{35} = 0.82 \text{ decodings.}$$

so per prefill step I evict $0.82 \cdot 4096 \simeq 3358$ tokens.

Question 0.12

Assuming each server produces:

$$\text{Prefills/s} = \frac{1}{0.9}$$

$$\text{Decodings/s} = \frac{32}{0.008 \cdot 512} = 7.81$$

Therefore we need to produce prefills at the same rate of the decoding servers:

$$N \cdot \frac{1}{0.9} = 7.81 \quad \Rightarrow \quad N = 7.029$$

So we need 7 more prefill servers than decoding servers.

Question 1:

We can use the formula from this chapter:

$$\text{Total FLOPs} = 2 \cdot M \cdot B$$

hence per token is:

$$\begin{aligned}\text{FLOPs / Token} &= 2 \cdot M \\ &= 2 \cdot 405e9 = 810e9\end{aligned}$$

If we are FLOPs bound a lower bound on N TPU v5e chips is:

$$t \geq \frac{2M}{N \cdot C} = \frac{810e9}{N \cdot 1.97e14}$$

If we are communication bound a lower bound is:

$$t \geq \frac{2BD}{M_x \cdot W_{IC1}} = \frac{2 \cdot B \cdot 16384}{M_x \cdot 9.0e10}$$

Question 2:

For LLaMa 3 - 8B we have $L = 32$,
 $D = 4096$, $F = 14336$, $N = 32$, $K = 8$,
 $H = 128$ and $V = 128256$.

Assuming int8 for everything we have:

$M \approx 8 \times 9$ Bytes.

$$\frac{KV \text{ Caches Size}}{S} = \frac{B \cdot 8 \cdot 2 \cdot L \cdot K \cdot H}{8} = 240 \cdot 2 \cdot 32 \cdot 8 \cdot 128 =$$

$$\approx 15.7 \text{ MB.}$$

For peak working activations we will assume:

$$I_{in} [B, S, D] \cdot W_{in} [D, F] \rightarrow Out [B, S, F]$$

hence per token $B \cdot F = 240 \cdot 14336 \text{ Bytes} \approx 3.4 \text{ MB}$
are used.

The smallest topology is a single TPU v5e.

Question 3:

For LLaMa 3-405B we have $L = 126$,
 $D = 16384$, $F = 53248$, $N = 128$, $K = 8$,
 $H = 128$ and $V = 128256$. Assume int8 weights
and bfloat16 FLOPs. Then we know that:

$$\begin{aligned} \text{KV cache size / Token} &= 2 \cdot L \cdot K \cdot H = 2 \cdot 126 \cdot 8 \cdot 128 \\ &\approx 258e3 \text{ Bytes} \\ &= 258 \text{ KB.} \end{aligned}$$

Assume $S = 8192$ as in previous exercises, then:

$$\begin{aligned} \text{Total Memory} &= M + B \cdot S \cdot 2 \cdot L \cdot K \cdot H = \\ &= 405e9 + B \cdot S \cdot 258e3 \end{aligned}$$

hence we need a slice vSe topology with at
least 408 GB of aggregated HBM (assuming
 $B=1$). Since HBM of TPU vSe is 16 GB we
need $1 \times 1 \cdot 16 \text{ GB} \not\geq 408 \text{ GB} \Leftrightarrow 1 \times 1 \not\geq 25.5$.

Note that a lower bound in latency is :

$$\frac{405e9}{|X| \cdot W_{HBM}} \leq \text{Step Time}$$

hence if we need to have a latency of $\leq 15 \text{ ms}$, we have to satisfy :

$$\frac{405e9}{0.015 \cdot 8.2e11} \leq |X| \Rightarrow |X| \geq 32.9$$

So we need more than 32 devices. We can consider a 8×8 TPU v5e slice, and use model parallelism. During prefill we will be memory bound since :

$$\frac{F}{2188} = 24 \neq |X| = 64$$

However we can improve our latency during generation.

Note that for a choice of B we are memory bound if :

$$T_{HBM \text{ comms}}(\text{int8}) = \frac{DF}{|X| \cdot W_{HBM}}$$

$$T_{ICI \text{ comms}}(\text{bfloat16}) = \frac{2BD}{W_{ICI}}$$

$$\frac{DF}{|X| \cdot W_{HBM}} > \frac{2BD}{W_{IC1}} \Leftrightarrow \frac{F \cdot W_{IC1}}{2 \cdot |X| \cdot W_{HBM}} > B$$

So then in order to be memory bound we need $B \leq 52.0$. At this batch size we are memory bound in the MLP. We can solve now for our max S length that still fits:

$$S \leq \frac{|X| \cdot HBM - M}{B \cdot 2 \cdot L \cdot K \cdot H} \simeq 46130$$

however we need S to be smaller for otherwise we will not hit the 15ms requirement.

$$\text{Step time} = \frac{52 \cdot S \cdot 258e3 + 405e4}{64 \cdot 8.2e11} \leq 0.015$$

$$S \leq \frac{64 \cdot 8.2e11 \cdot 0.015 - 405e4}{52 \cdot 258e3} \simeq 28488$$

Therefore with $B = 52$ and $S = 28488$ we have :

$$\text{Step Time} = 15 \text{ ms}$$

and a throughput of :

$$\text{Tokens/s} = \frac{B}{0.015} = \frac{52}{0.015} = 66.$$